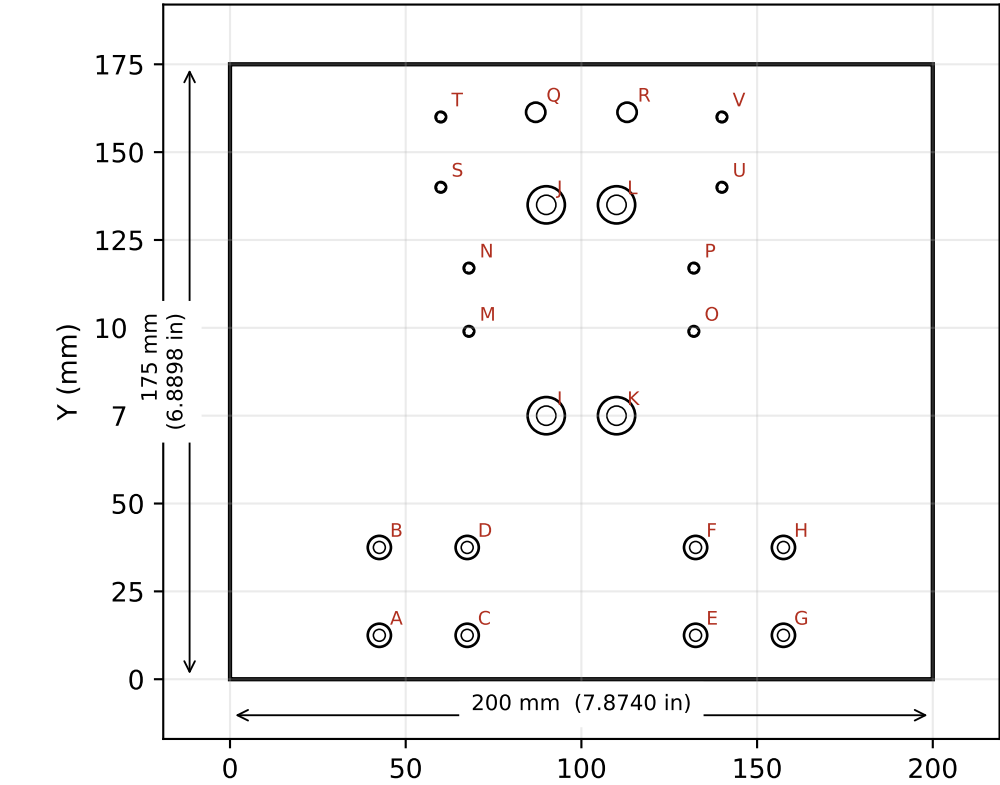


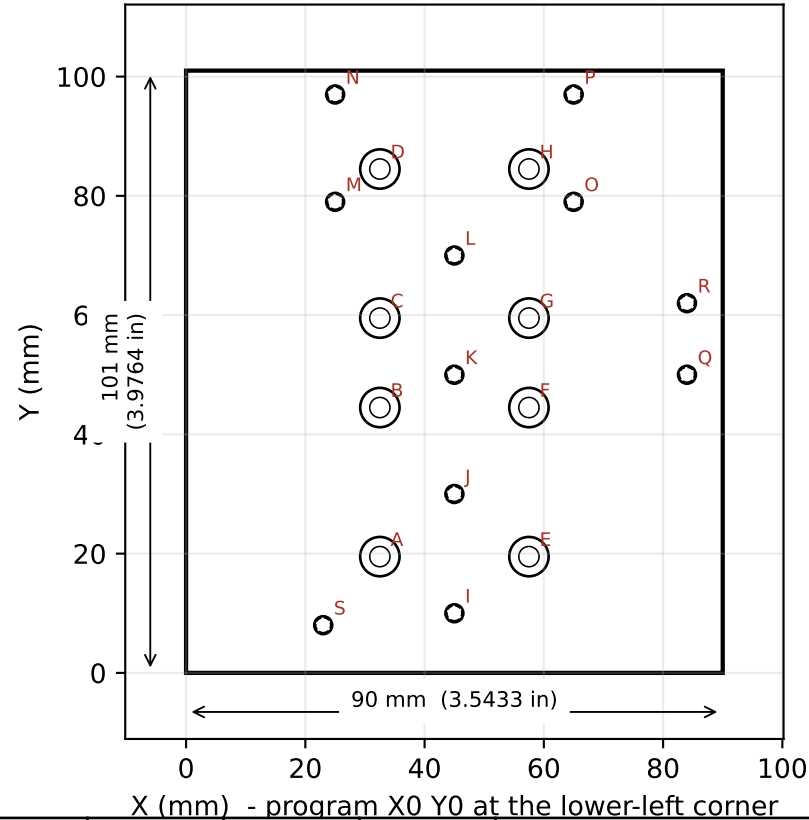
PEN PLOTTER — Gantry end plate · OP1 outer face



PART	Gantry end plate
ID	END_PLATE
SETUP	OP1 outer face
PROGRAM	O60011 (O60011_END_PLATE_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Countersinks face OUTWARD (away from the rail). Same part both sides.

X (mm) - program X0 Y0 at the lower-left corner							
ID	X in	Y in	X mm	Y mm	Feature	Purpose	
A	1.6732	0.4921	42.50	12.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
B	1.6732	1.4764	42.50	37.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
C	2.6575	0.4921	67.50	12.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
D	2.6575	1.4764	67.50	37.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
E	5.2165	0.4921	132.50	12.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
F	5.2165	1.4764	132.50	37.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
G	6.2008	0.4921	157.50	12.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
H	6.2008	1.4764	157.50	37.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block	
I	3.5433	2.9528	90.00	75.00	$\varnothing 5.5$ THRU, CSK $\varnothing 10.6 \times 90^\circ$ (M5 FHCS)	beam end	
J	3.5433	5.3150	90.00	135.00	$\varnothing 5.5$ THRU, CSK $\varnothing 10.6 \times 90^\circ$ (M5 FHCS)	beam end	
K	4.3307	2.9528	110.00	75.00	$\varnothing 5.5$ THRU, CSK $\varnothing 10.6 \times 90^\circ$ (M5 FHCS)	beam end	
L	4.3307	5.3150	110.00	135.00	$\varnothing 5.5$ THRU, CSK $\varnothing 10.6 \times 90^\circ$ (M5 FHCS)	beam end	
M	2.6772	3.8976	68.00	99.00	M3x0.5 THRU (drill $\varnothing 2.5$)	Y belt clamp bar	
N	2.6772	4.6063	68.00	117.00	M3x0.5 THRU (drill $\varnothing 2.5$)	Y belt clamp bar	
O	5.1969	3.8976	132.00	99.00	M3x0.5 THRU (drill $\varnothing 2.5$)	Y belt clamp bar	
P	5.1969	4.6063	132.00	117.00	M3x0.5 THRU (drill $\varnothing 2.5$)	Y belt clamp bar	
Q	3.4252	6.3524	87.00	161.35	$\varnothing 5.5$ THRU (M5 clr)	X belt tension screw	
R	4.4488	6.3524	113.00	161.35	$\varnothing 5.5$ THRU (M5 clr)	X belt tension screw	
S	2.3622	5.5118	60.00	140.00	M3x0.5 THRU (drill $\varnothing 2.5$)	cable-chain bracket	
T	2.3622	6.2992	60.00	160.00	M3x0.5 THRU (drill $\varnothing 2.5$)	cable-chain bracket	
U	5.5118	5.5118	140.00	140.00	M3x0.5 THRU (drill $\varnothing 2.5$)	cable-chain bracket	
V	5.5118	6.2992	140.00	160.00	M3x0.5 THRU (drill $\varnothing 2.5$)	cable-chain bracket	

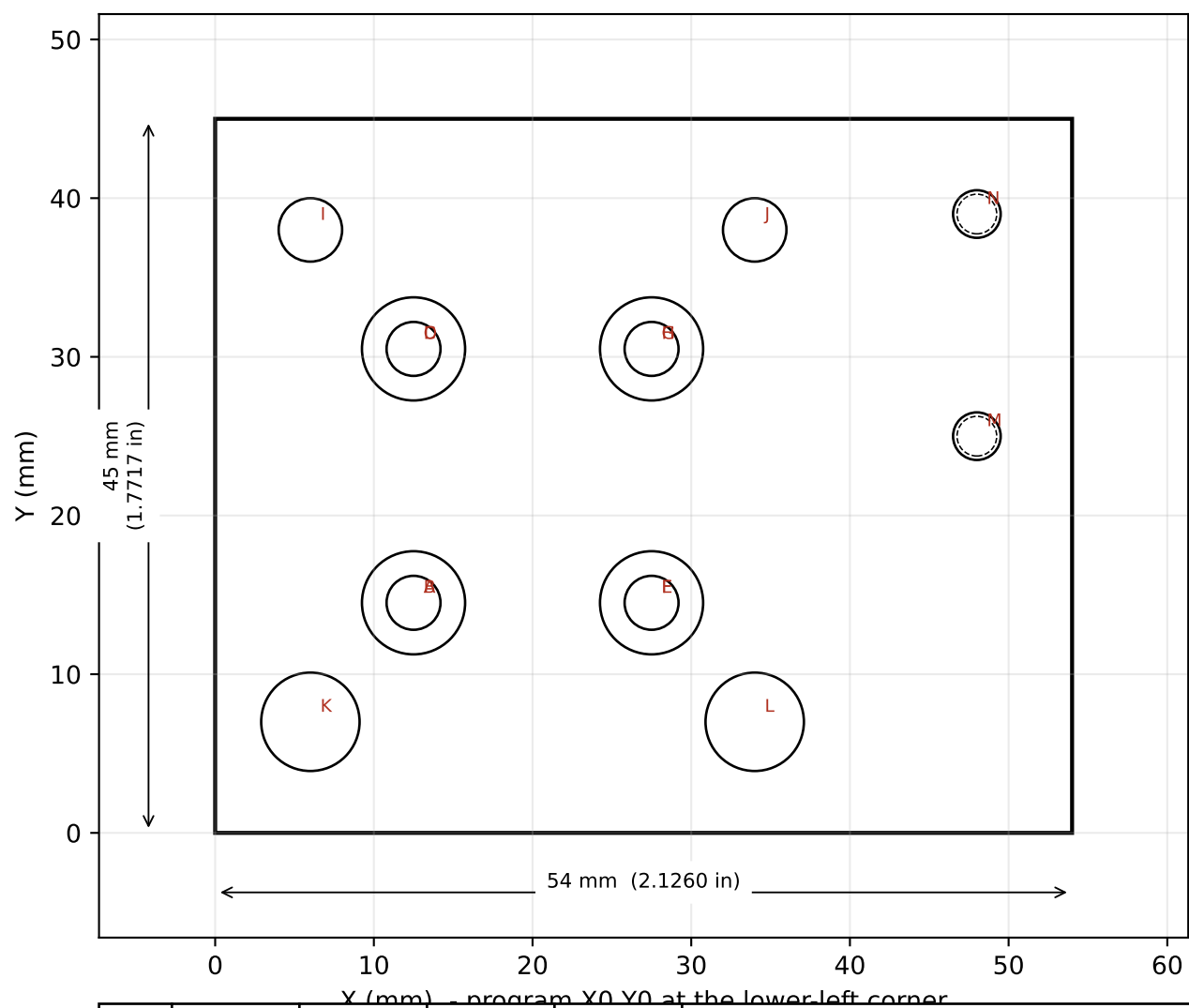
PEN PLOTTER — Pen carriage plate · OP1 front face



PART	Pen carriage plate
ID	PEN_CARRIAGE
SETUP	OP1 front face
PROGRAM	O60021 (O60021_PEN_CARRIAGE_OP1.nc)
QTY	1
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Countersinks face FORWARD (away from the beam).

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	1.2795	0.7677	32.50	19.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
B	1.2795	1.7520	32.50	44.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
C	1.2795	2.3425	32.50	59.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
D	1.2795	3.3268	32.50	84.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
E	2.2638	0.7677	57.50	19.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
F	2.2638	1.7520	57.50	44.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
G	2.2638	2.3425	57.50	59.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
H	2.2638	3.3268	57.50	84.50	$\varnothing 3.4$ THRU, CSK $\varnothing 6.6 \times 90^\circ$ (M3 FHCS)	MGN15H block
I	1.7717	0.3937	45.00	10.00	M3x0.5 THRU (drill $\varnothing 2.5$)	MGN9 rail
J	1.7717	1.1811	45.00	30.00	M3x0.5 THRU (drill $\varnothing 2.5$)	MGN9 rail
K	1.7717	1.9685	45.00	50.00	M3x0.5 THRU (drill $\varnothing 2.5$)	MGN9 rail
L	1.7717	2.7559	45.00	70.00	M3x0.5 THRU (drill $\varnothing 2.5$)	MGN9 rail
M	0.9843	3.1102	25.00	79.00	M3x0.5 THRU (drill $\varnothing 2.5$)	X belt clamp bar
N	0.9843	3.8189	25.00	97.00	M3x0.5 THRU (drill $\varnothing 2.5$)	X belt clamp bar
O	2.5591	3.1102	65.00	79.00	M3x0.5 THRU (drill $\varnothing 2.5$)	X belt clamp bar
P	2.5591	3.8189	65.00	97.00	M3x0.5 THRU (drill $\varnothing 2.5$)	X belt clamp bar
Q	3.3071	1.9685	84.00	50.00	M3x0.5 THRU (drill $\varnothing 2.5$)	servo bracket
R	3.3071	2.4409	84.00	62.00	M3x0.5 THRU (drill $\varnothing 2.5$)	servo bracket
S	0.9055	0.3150	23.00	8.00	M3x0.5 THRU (drill $\varnothing 2.5$)	dock stop

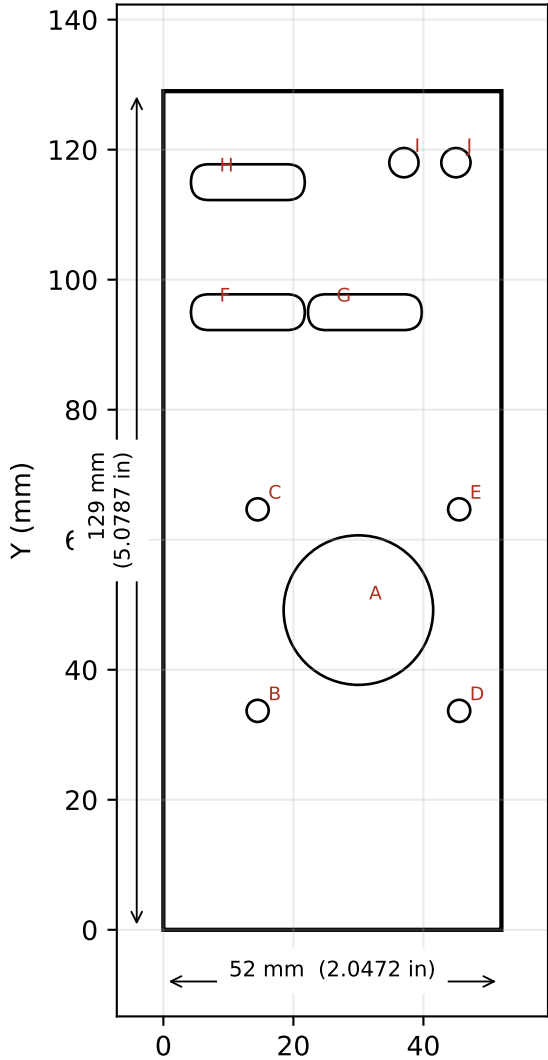
PEN PLOTTER — Pen dock · OP1 front face



PART	Pen dock
ID	PEN_DOCK
SETUP	OP1 front face
PROGRAM	O60031 (O60031_PEN_DOCK_OP1.nc)
QTY	1
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature		Purpose
A	0.4921	0.5709	12.50	14.50	$\varnothing 3.4$ THRU (M3 clr)		MGN9H block, M3x5 SHCS
B	0.4921	0.5709	12.50	14.50	$\varnothing 6.5 \times 4$ DP POCKET		M3 SHCS counterbore
C	0.4921	1.2008	12.50	30.50	$\varnothing 3.4$ THRU (M3 clr)		MGN9H block, M3x5 SHCS
D	0.4921	1.2008	12.50	30.50	$\varnothing 6.5 \times 4$ DP POCKET		M3 SHCS counterbore
E	1.0827	0.5709	27.50	14.50	$\varnothing 3.4$ THRU (M3 clr)		MGN9H block, M3x5 SHCS
F	1.0827	0.5709	27.50	14.50	$\varnothing 6.5 \times 4$ DP POCKET		M3 SHCS counterbore
G	1.0827	1.2008	27.50	30.50	$\varnothing 3.4$ THRU (M3 clr)		MGN9H block, M3x5 SHCS
H	1.0827	1.2008	27.50	30.50	$\varnothing 6.5 \times 4$ DP POCKET		M3 SHCS counterbore
I	0.2362	1.4961	6.00	38.00	$\varnothing 4$ H7 REAM THRU		4 mm dowel, press fit + retaining compound
J	1.3386	1.4961	34.00	38.00	$\varnothing 4$ H7 REAM THRU		4 mm dowel, press fit + retaining compound
K	0.2362	0.2756	6.00	7.00	$\varnothing 6.2 \times 3.2$ DP POCKET		6 x 3 mm N52 magnet, epoxy
L	1.3386	0.2756	34.00	7.00	$\varnothing 6.2 \times 3.2$ DP POCKET		6 x 3 mm N52 magnet, epoxy
M	1.8898	0.9843	48.00	25.00	M3x0.5 THRU (drill $\varnothing 2.5$)		lift tab (M3x10 SHCS, head forward)
N	1.8898	1.5354	48.00	39.00	M3x0.5 THRU (drill $\varnothing 2.5$)		lift tab (M3x10 SHCS, head forward)

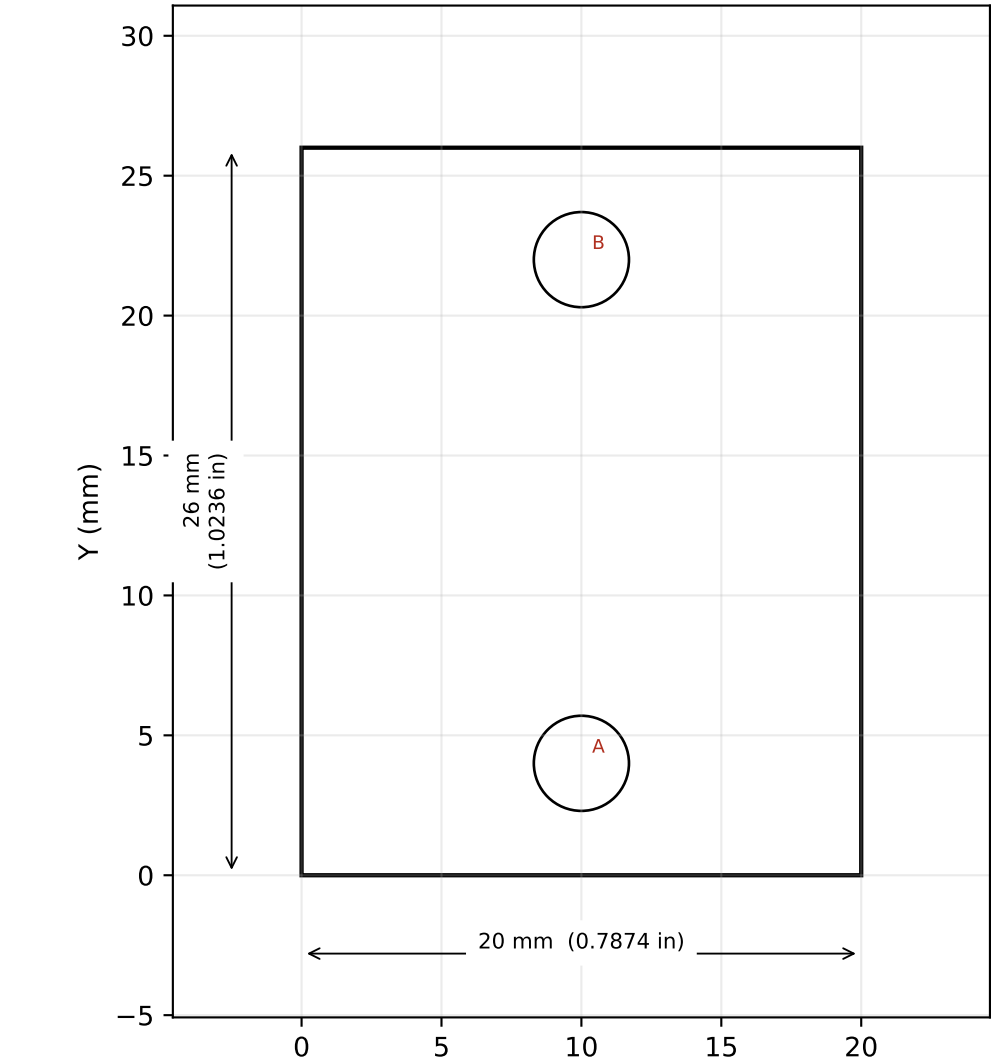
PEN PLOTTER — X motor / idler plate · OP1 top face



PART	X motor / idler plate
ID	X_MOTOR_PLATE
SETUP	OP1 top face
PROGRAM	O60051 (O60051_X_MOTOR_PLATE_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Right side carries the X motor; the left one is the same part flipped over and carries the idler.

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	1.1811	1.9360	30.00	49.17	$\varnothing 23$ THRU (helical)	NEMA17 pilot / idler bolt clearance
B	0.5709	1.3258	14.50	33.67	$\varnothing 3.4$ THRU (M3 clr)	NEMA17 screws
C	0.5709	2.5463	14.50	64.67	$\varnothing 3.4$ THRU (M3 clr)	NEMA17 screws
D	1.7913	1.3258	45.50	33.67	$\varnothing 3.4$ THRU (M3 clr)	NEMA17 screws
E	1.7913	2.5463	45.50	64.67	$\varnothing 3.4$ THRU (M3 clr)	NEMA17 screws
F	0.2756	3.7402	7.00	95.00	SLOT 5.5 WIDE x 17.5 LONG THRU	beam T-nut, slides for X belt tension
G	0.9843	3.7402	25.00	95.00	SLOT 5.5 WIDE x 17.5 LONG THRU	beam T-nut
H	0.2756	4.5276	7.00	115.00	SLOT 5.5 WIDE x 17.5 LONG THRU	beam T-nut
I	1.4567	4.6457	37.00	118.00	$\varnothing 4.5$ THRU (M4 clr)	tension block (idler side)
J	1.7717	4.6457	45.00	118.00	$\varnothing 4.5$ THRU (M4 clr)	tension block (idler side)

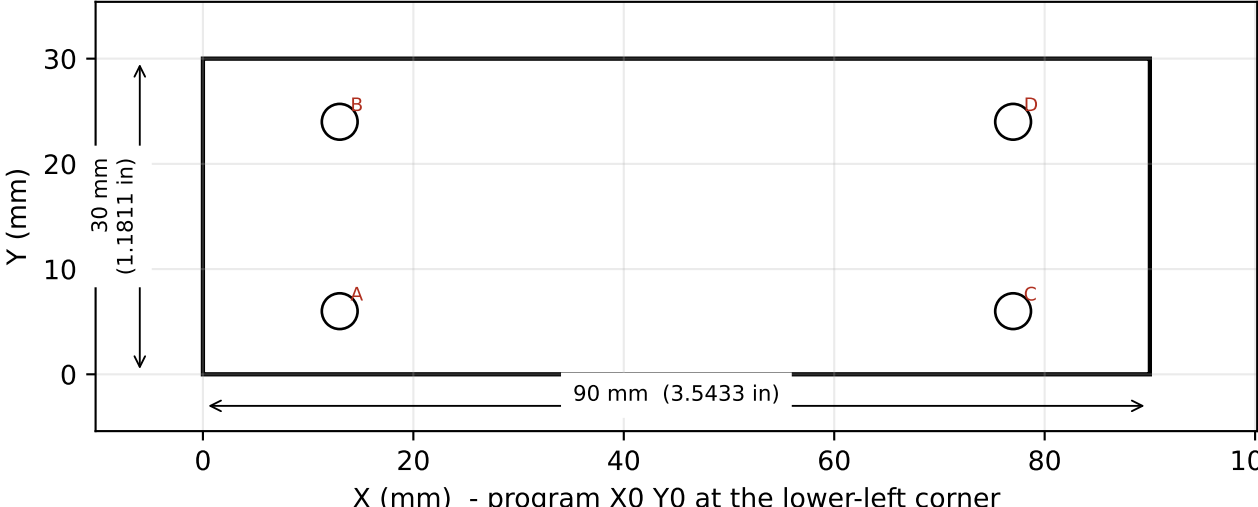
PEN PLOTTER — X belt clamp bar · OP1



PART	X belt clamp bar
ID	X_CLAMP_BAR
SETUP	OP1
PROGRAM	O60061 (O60061_X_CLAMP_BAR_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ±0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.3937	0.1575	10.00	4.00	Ø3.4 THRU (M3 clr)	
B	0.3937	0.8661	10.00	22.00	Ø3.4 THRU (M3 clr)	

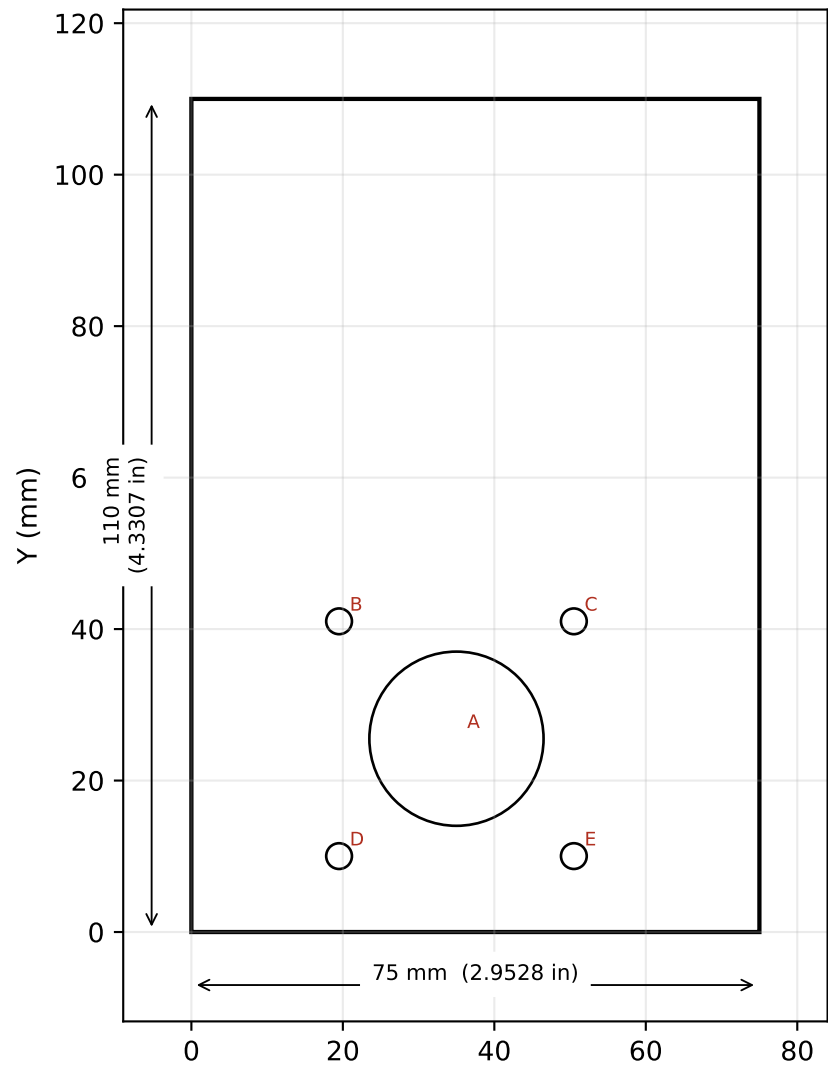
PEN PLOTTER — Y belt clamp bar · OP1



PART	Y belt clamp bar
ID	Y_CLAMP_BAR
SETUP	OP1
PROGRAM	O60071 (O60071_Y_CLAMP_BAR_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.5118	0.2362	13.00	6.00	Ø3.4 THRU (M3 clr)	
B	0.5118	0.9449	13.00	24.00	Ø3.4 THRU (M3 clr)	
C	3.0315	0.2362	77.00	6.00	Ø3.4 THRU (M3 clr)	
D	3.0315	0.9449	77.00	24.00	Ø3.4 THRU (M3 clr)	

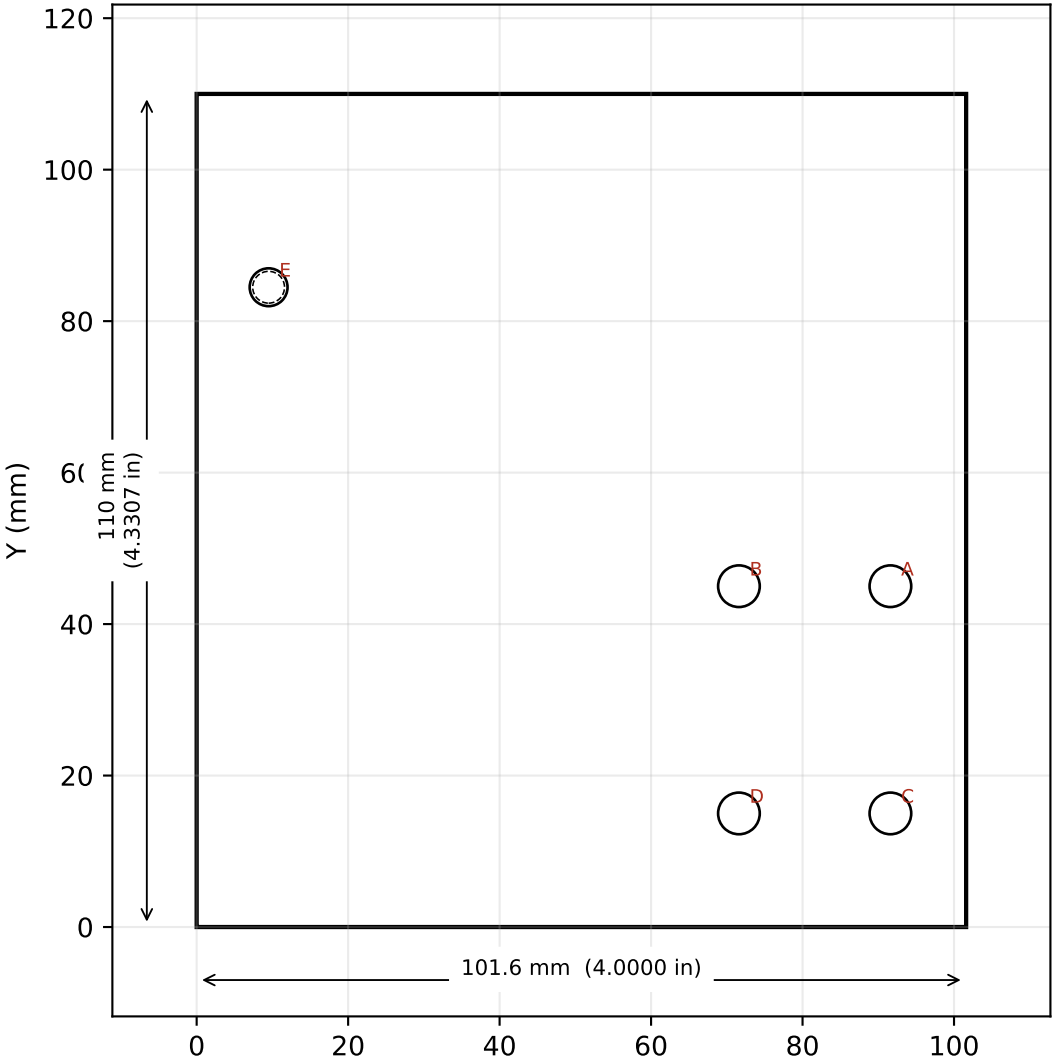
PEN PLOTTER — Y motor/idler bracket A · OP1 leg B (horizontal leg, top face)



PART	Y motor/idler bracket A
ID	Y_BRACKET_A
SETUP	OP1 leg B (horizontal leg, top face)
PROGRAM	O60081 (O60081_Y_BRACKET_A_OP1.nc)
QTY	2
MATERIAL	4 x 4 x 1/4 in 6061-T6 angle, sawn 110 long, leg B trimmed to 75
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Grip by the two sawn ends - 110 mm apart along y. leg b flat on parallels, outer face up
ORIENT	Corner at x0 - leg a hangs down past the x- end of the jaws
TOLERANCE	holes ±0.05 mm; break all edges 0.3 mm
NOTES	Bolts to the frame end; doubles as the corner joint.

ID	X in	Y in	X mm	Y mm	Feature		Purpose
A	1.3780	1.0049	35.00	25.52	Ø23 THRU (helical)	NEMA17 pilot (motor end) / idler bolt clearance (idler end)	
B	0.7677	1.6152	19.50	41.02	Ø3.4 THRU (M3 clr)		NEMA17 screws
C	1.9882	1.6152	50.50	41.02	Ø3.4 THRU (M3 clr)		NEMA17 screws
D	0.7677	0.3947	19.50	10.02	Ø3.4 THRU (M3 clr)		NEMA17 screws
E	1.9882	0.3947	50.50	10.02	Ø3.4 THRU (M3 clr)		NEMA17 screws

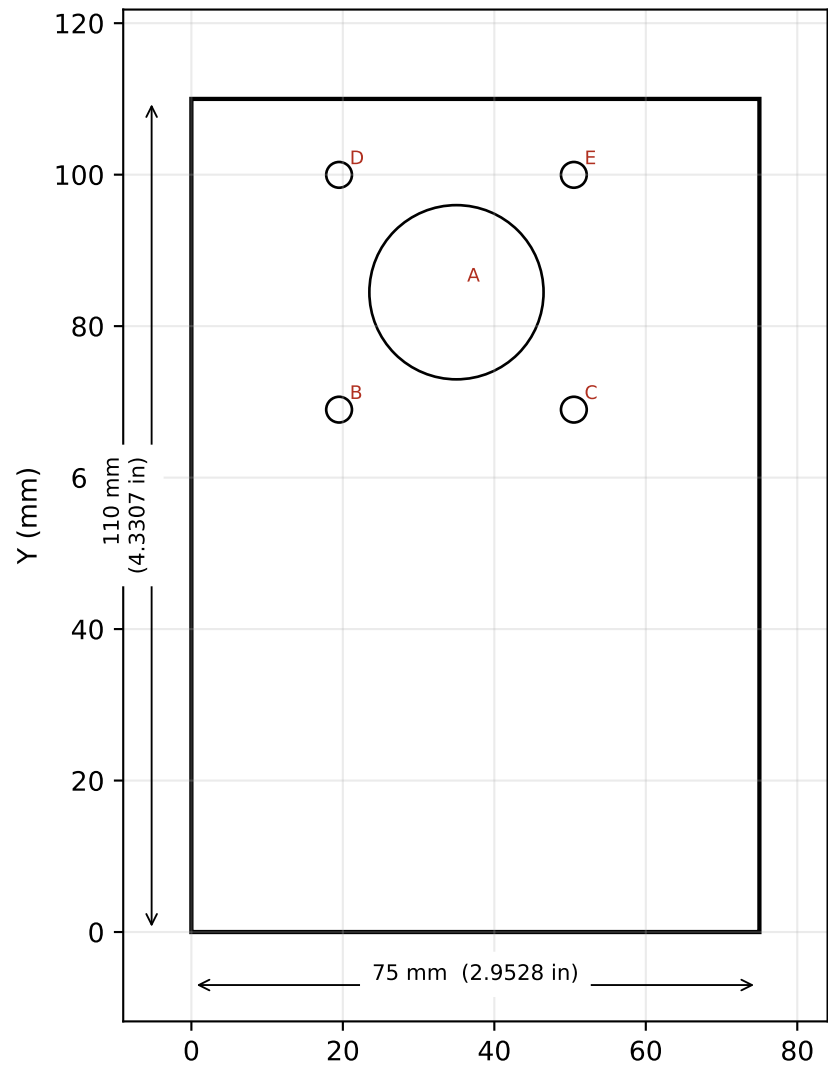
PEN PLOTTER — Y motor/idler bracket A · OP2 leg A (vertical leg, outer face)



PART	Y motor/idler bracket A
ID	Y_BRACKET_A
SETUP	OP2 leg A (vertical leg, outer face)
PROGRAM	O60082 (O60082_Y_BRACKET_A_OP2.nc)
QTY	2
MATERIAL	4 x 4 x 1/4 in 6061-T6 angle, sawn 110 long, leg B trimmed to 75
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Grip by the sawn ends. leg a flat on parallels, frame-side face up
ORIENT	Corner at x0 - leg b hangs down past the x- end of the jaws
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Bolts to the frame end; doubles as the corner joint.

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	3.6063	1.7717	91.60	45.00	$\varnothing 5.5$ THRU (M5 clr)	frame end tap
B	2.8189	1.7717	71.60	45.00	$\varnothing 5.5$ THRU (M5 clr)	frame end tap
C	3.6063	0.5906	91.60	15.00	$\varnothing 5.5$ THRU (M5 clr)	cross-member T-nut
D	2.8189	0.5906	71.60	15.00	$\varnothing 5.5$ THRU (M5 clr)	cross-member T-nut
E	0.3740	3.3258	9.50	84.48	M5x0.8 THRU (drill $\varnothing 4.2$)	idler-carrier jack screw (drill through fillet)

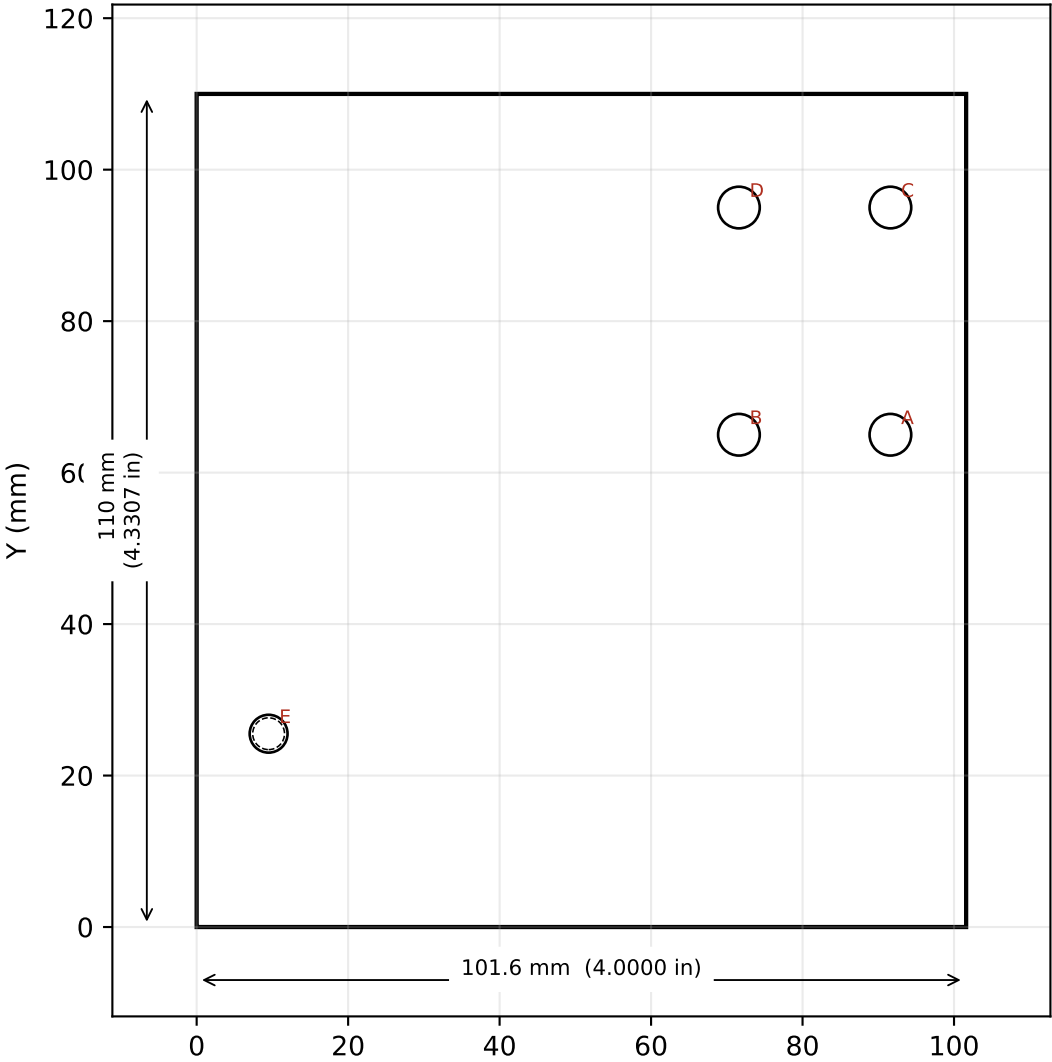
PEN PLOTTER — Y motor/idler bracket B (mirror of A) · OP1 leg B (horizontal leg, top face)



PART	Y motor/idler bracket B (mirror of A)
ID	Y_BRACKET_B
SETUP	OP1 leg B (horizontal leg, top face)
PROGRAM	O60091 (O60091_Y_BRACKET_B_OP1.nc)
QTY	2
MATERIAL	4 x 4 x 1/4 in 6061-T6 angle, sawn 110 long, leg B trimmed to 75
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Grip by the two sawn ends - 110 mm apart along y. leg b flat on parallels, outer face up
ORIENT	Corner at x0 - leg a hangs down past the x- end of the jaws
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Bolts to the frame end; doubles as the corner joint.

ID	X in	Y in	X mm	Y mm	Feature		Purpose
A	1.3780	3.3258	35.00	84.48	Ø23 THRU (helical)	NEMA17 pilot (motor end) / idler bolt clearance (idler end)	
B	0.7677	2.7156	19.50	68.98	Ø3.4 THRU (M3 clr)		NEMA17 screws
C	1.9882	2.7156	50.50	68.98	Ø3.4 THRU (M3 clr)		NEMA17 screws
D	0.7677	3.9360	19.50	99.98	Ø3.4 THRU (M3 clr)		NEMA17 screws
E	1.9882	3.9360	50.50	99.98	Ø3.4 THRU (M3 clr)		NEMA17 screws

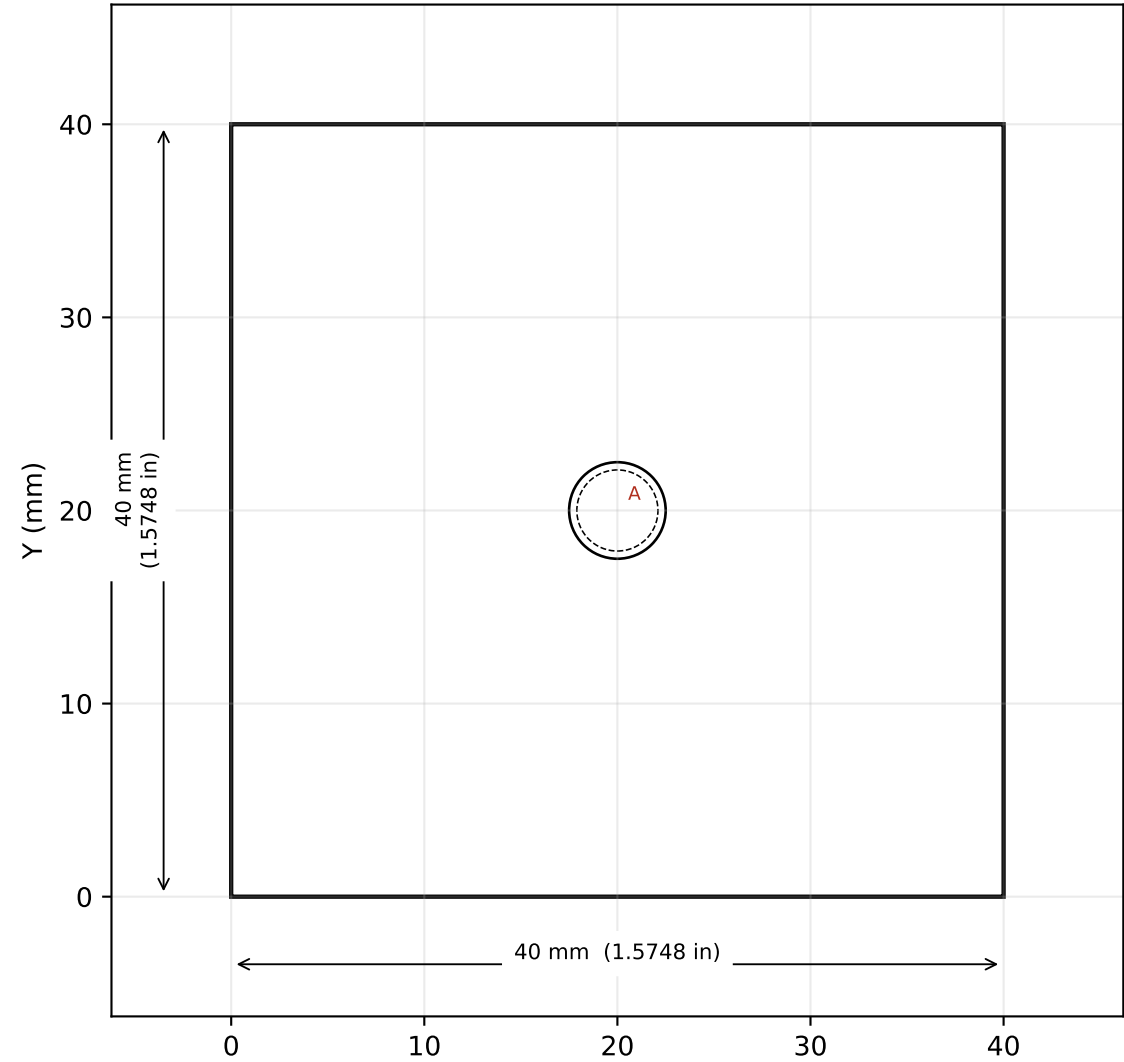
PEN PLOTTER — Y motor/idler bracket B (mirror of A) · OP2 leg A (vertical leg, outer face)



PART	Y motor/idler bracket B (mirror of A)
ID	Y_BRACKET_B
SETUP	OP2 leg A (vertical leg, outer face)
PROGRAM	O60092 (O60092_Y_BRACKET_B_OP2.nc)
QTY	2
MATERIAL	4 x 4 x 1/4 in 6061-T6 angle, sawn 110 long, leg B trimmed to 75
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Grip by the sawn ends. leg a flat on parallels, frame-side face up
ORIENT	Corner at x0 - leg b hangs down past the x- end of the jaws
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Bolts to the frame end; doubles as the corner joint.

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	3.6063	2.5591	91.60	65.00	$\varnothing 5.5$ THRU (M5 clr)	frame end tap
B	2.8189	2.5591	71.60	65.00	$\varnothing 5.5$ THRU (M5 clr)	frame end tap
C	3.6063	3.7402	91.60	95.00	$\varnothing 5.5$ THRU (M5 clr)	cross-member T-nut
D	2.8189	3.7402	71.60	95.00	$\varnothing 5.5$ THRU (M5 clr)	cross-member T-nut
E	0.3740	1.0049	9.50	25.52	M5x0.8 THRU (drill $\varnothing 4.2$)	idler-carrier jack screw (drill through fillet)

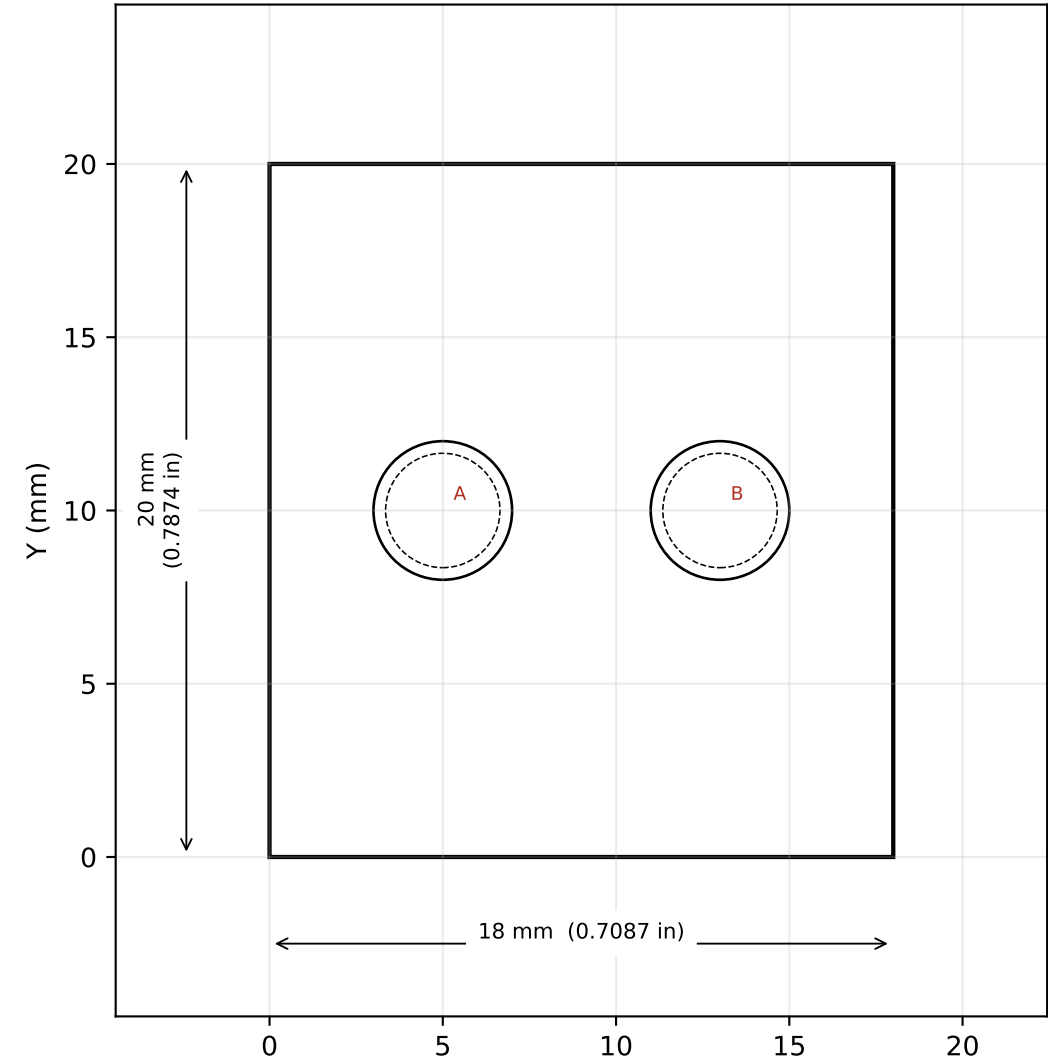
PEN PLOTTER — Y idler carrier · OP1



PART	Y idler carrier
ID	Y_IDLER_CARRIER
SETUP	OP1
PROGRAM	O60101 (O60101_Y_IDLER_CARRIER_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.7874	0.7874	20.00	20.00	M5x0.8 THRU (drill Ø4.2)	idler axle bolt (M5 SHCS from above)

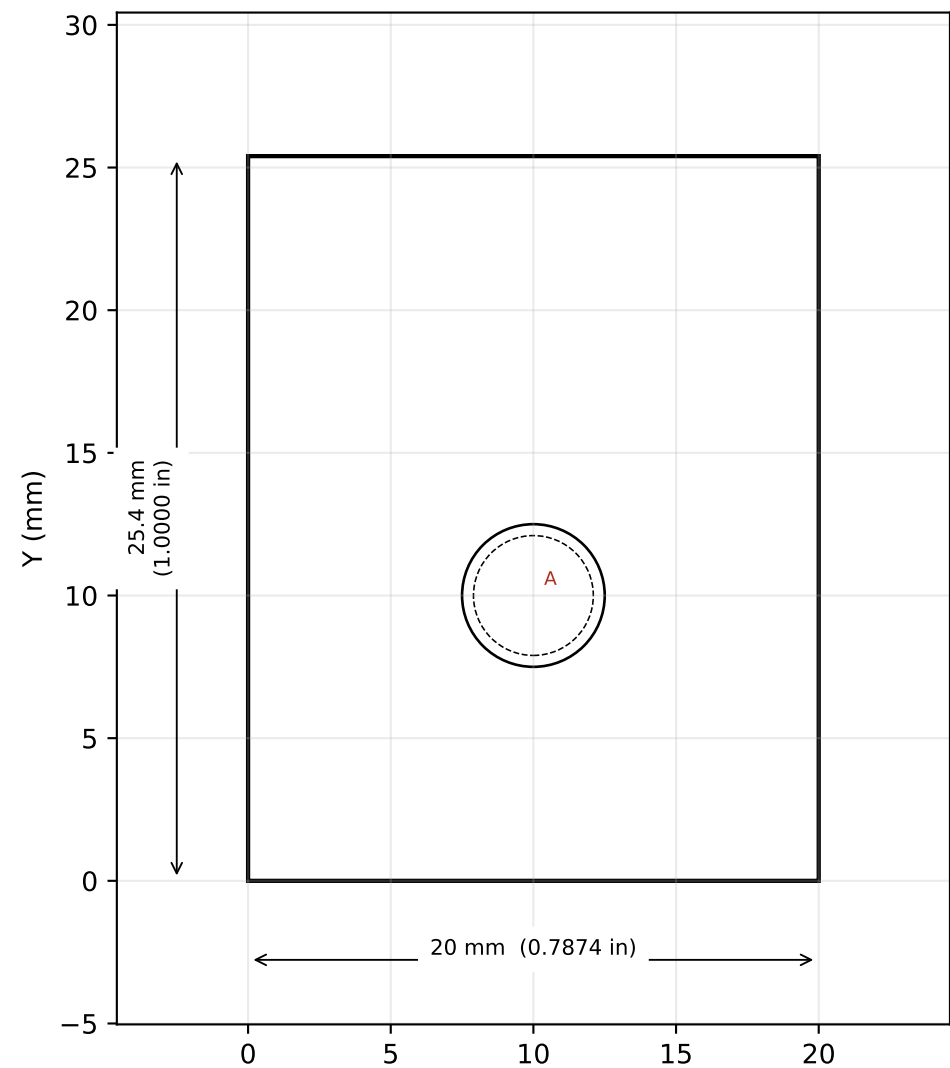
PEN PLOTTER — X belt tension block · OP1 bottom face



PART	X belt tension block
ID	X_TENSION_BLOCK
SETUP	OP1 bottom face
PROGRAM	O60111 (O60111_X_TENSION_BLOCK_OP1.nc)
QTY	1
MATERIAL	1 x 1 in 6061 bar, sawn 20 long
THICKNESS	25.4 mm (1.000 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.1969	0.3937	5.00	10.00	M4x0.7 x 10 DP (drill Ø3.3)	
B	0.5118	0.3937	13.00	10.00	M4x0.7 x 10 DP (drill Ø3.3)	

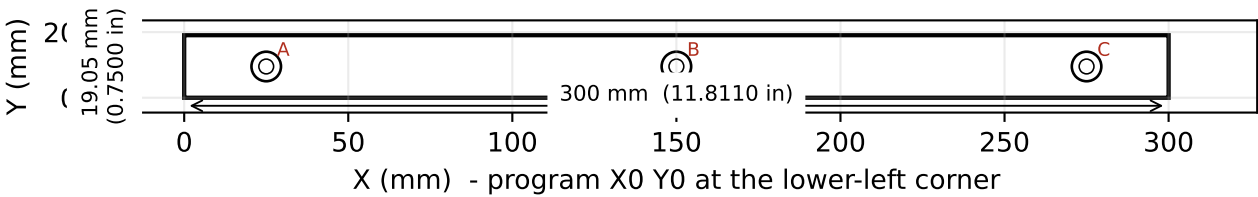
PEN PLOTTER — X belt tension block · OP2 end face (faces the left end plate)



PART	X belt tension block
ID	X_TENSION_BLOCK
SETUP	OP2 end face (faces the left end plate)
PROGRAM	O60112 (O60112_X_TENSION_BLOCK_OP2.nc)
QTY	1
MATERIAL	1 x 1 in 6061 bar, sawn 20 long
THICKNESS	18 mm (0.709 in)
FIXTURE	Stand the block on end in the vise, end face up
ORIENT	X along the 20 mm side, y0 = bottom face (op1 face) edge
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.3937	0.3937	10.00	10.00	M5x0.8 x 15 DP (drill Ø4.2)	

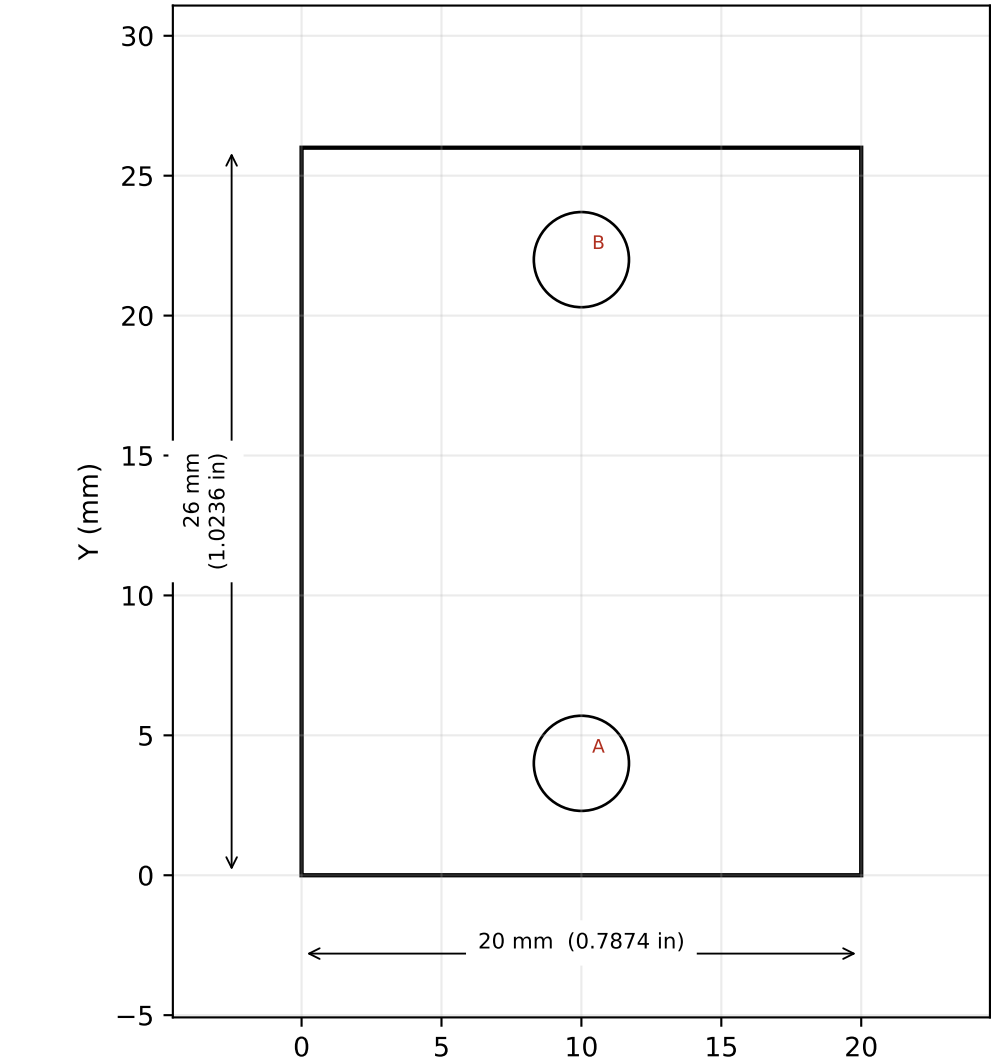
PEN PLOTTER — Paper corner stop bar · OP1



PART	Paper corner stop bar
ID	PAPER_STOP
SETUP	OP1
PROGRAM	O60121 (O60121_PAPER_STOP_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate, sawn strip 300 x 19.05
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ±0.05 mm; break all edges 0.3 mm
NOTES	-

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.9843	0.3750	25.00	9.53	Ø4.5 THRU, CSK Ø9.0 x 90° (#8 FH wood screw)	#8 flat head wood screw, 90 deg csk to 9.0
B	5.9055	0.3750	150.00	9.53	Ø4.5 THRU, CSK Ø9.0 x 90° (#8 FH wood screw)	#8 flat head wood screw, 90 deg csk to 9.0
C	10.8268	0.3750	275.00	9.53	Ø4.5 THRU, CSK Ø9.0 x 90° (#8 FH wood screw)	#8 flat head wood screw, 90 deg csk to 9.0

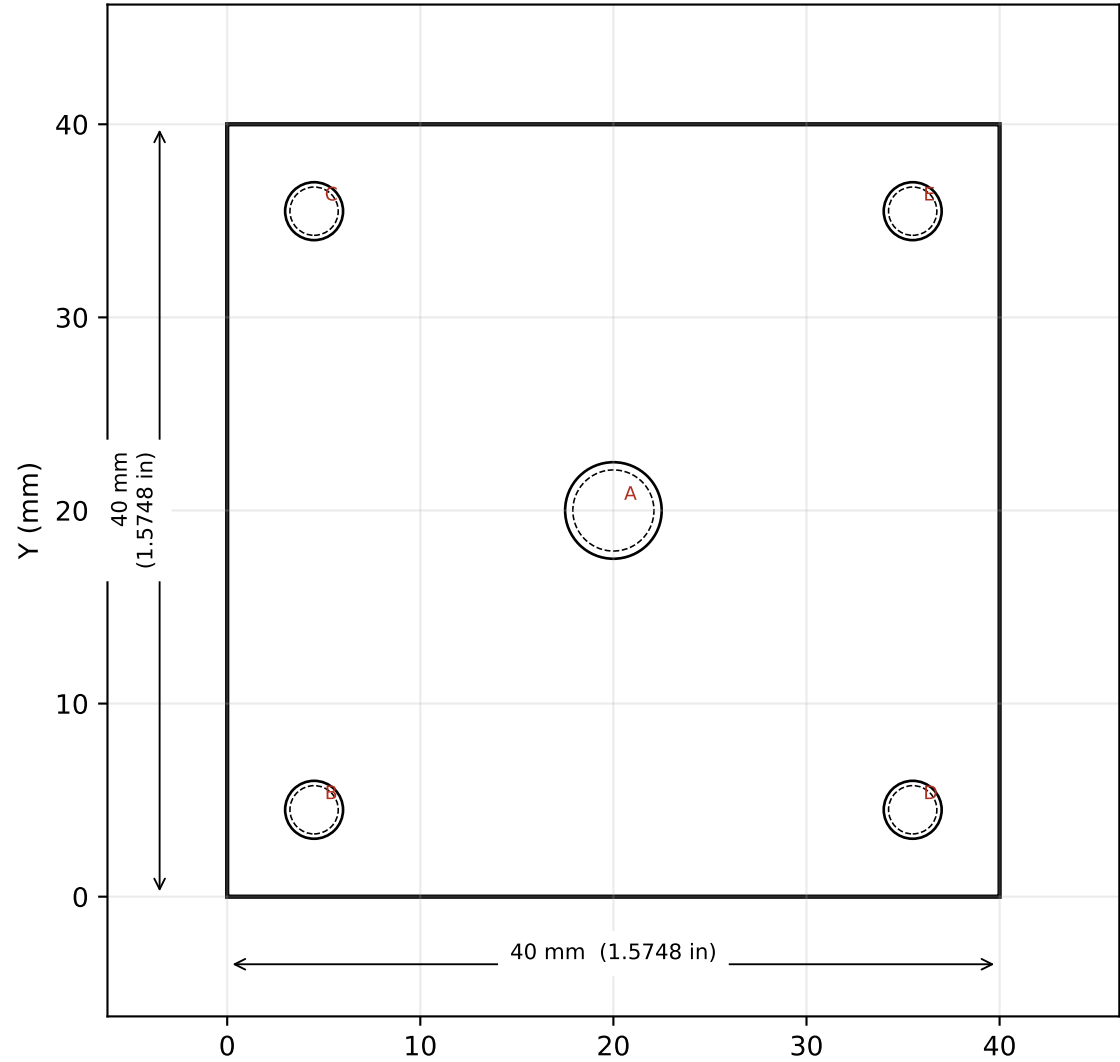
PEN PLOTTER — X belt spacer · OP1



PART	X belt spacer
ID	X_BELT_SPACER
SETUP	OP1
PROGRAM	O60131 (O60131_X_BELT_SPACER_OP1.nc)
QTY	2
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Same blank and holes as the X clamp bar: run the clamp bar program 4 times and use 2 as spacers.

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.3937	0.1575	10.00	4.00	Ø3.4 THRU (M3 clr)	
B	0.3937	0.8661	10.00	22.00	Ø3.4 THRU (M3 clr)	

PEN PLOTTER — X idler carrier · OP1



PART	X idler carrier
ID	X_IDLER_CARRIER
SETUP	OP1
PROGRAM	O60141 (O60141_X_IDLER_CARRIER_OP1.nc)
QTY	1
MATERIAL	1/4" 6061-T651 plate
THICKNESS	6.35 mm (0.250 in)
FIXTURE	Squared blank on parallels, grip the long edges, face up
ORIENT	Long side along x
TOLERANCE	holes ± 0.05 mm; break all edges 0.3 mm
NOTES	Sits on top of the left X plate. Machined face UP. Without it the X idler has no axle.

ID	X in	Y in	X mm	Y mm	Feature	Purpose
A	0.7874	0.7874	20.00	20.00	M5x0.8 THRU (drill Ø4.2)	idler axle bolt (M5 SHCS from below)
B	0.1772	0.1772	4.50	4.50	M3x0.5 THRU (drill Ø2.5)	idler plate NEMA17 clearance hole (M3x12 from below)
C	0.1772	1.3976	4.50	35.50	M3x0.5 THRU (drill Ø2.5)	idler plate NEMA17 clearance hole (M3x12 from below)
D	1.3976	0.1772	35.50	4.50	M3x0.5 THRU (drill Ø2.5)	idler plate NEMA17 clearance hole (M3x12 from below)
E	1.3976	1.3976	35.50	35.50	M3x0.5 THRU (drill Ø2.5)	idler plate NEMA17 clearance hole (M3x12 from below)